Learning Contact Residuals for In-Hand Manipulation Transfer

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Abstract: Dexterous in-hand manipulation relies on contact-rich dynamics that are difficult to simulate accurately. Existing work such as system identification of simulator parameters and dynamics randomization is constrained by the expressiveness of the simulator, while hand-focused learned dynamics can leave object contact effects implicit. We introduce an object-centric neural residual model within a real-to-sim-to-real framework that learns contact-conditioned corrections from target trajectories and fine-tunes policies in the augmented simulator. Preliminary sim-to-sim experiments show improved transfer across objects, tasks, and hand embodiments.

Keywords: Residual Dynamics, Sim2Real, Hybrid Simulation, Dexterous Manipulation

1 Introduction

Dexterous manipulation coordinates finger motion and changing contacts for skills from in-hand reorientation to tool use [1, 2]. Simulation offers scalable policy training, but inaccurate friction and contact transitions [3] allow policies to exploit interactions that fail to transfer.

Existing work such as system identification (SysID) calibrates simulator parameters to observations [4, 5], while domain randomization (DR) trains policies under parameter variation [6, 7]. HORA [8] adapts policies from proprioceptive history; RotateIt [9] adds visuotactile feedback for multi-axis rotation, and AnyRotate [10] uses contact pose and force feedback across hand orientations. These approaches retain the simulator's model structure, so discrepancies beyond its expressiveness can persist. Grounding simulation in target experience improves transfer [11]; learned dynamics corrections extend this approach beyond nominal models [12, 13]. For dexterity, DexNDM [14] learns joint dynamics to train a deployed action residual, leaving the relationship between contacts and object motion implicit. QuasiSim [15] explicitly learns contact residuals, but targets human-demonstration tracking through a physics curriculum.

We introduce an object-centric neural residual dynamics model for policy transfer. (Fig. 1) We use a surface-anchor field to explicitly condition object momentum corrections on contact and geometry, extending nominal physics beyond parameter changes. Paired replay of target trajectories supervises the residual; it then stays frozen during policy fine-tuning, and only the policy transfers.

Our contributions are:

1. A real-to-sim-to-real framework that augments nominal physics with learned residual dynamics for in-hand manipulation policy transfer.
2. An object-centric residual model that explicitly maps surface geometry and contact features to rotation-equivariant momentum corrections.
3. A broad sim-to-sim evidence of improved transfer and generalization across two hands, multiple objects, and various tasks.

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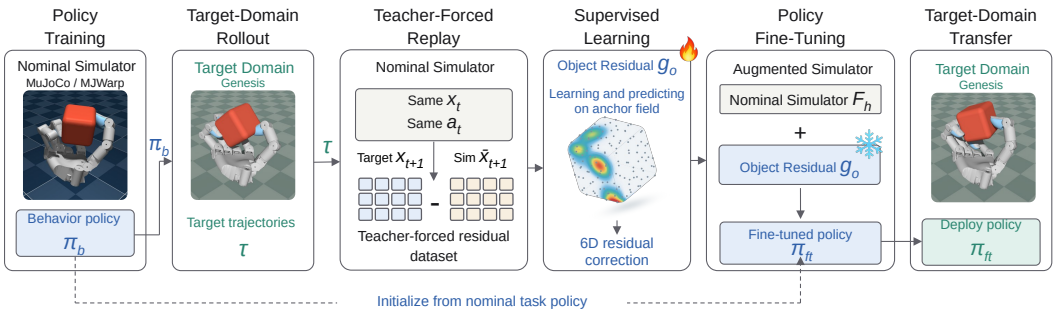


Figure 1: **Method Overview.** Paired replay of target trajectories from Genesis [16] provides momentum labels $\mathbf{y}_t$ for the object residual g_o. We then freeze g_o for policy fine-tuning in MuJoCo/MJWarp [17, 18] and deploy π_{ft} back in Genesis.

2 Method

We learn a contact-conditioned object residual g_o from paired replay of target trajectories. With g_o frozen, the augmented simulator supports policy fine-tuning and transfer (Fig. 1).

Behavior rollout and residual label. We use an in-hand reorientation behavior policy π_b to elicit active finger and palm contacts with the object. We train π_b with PPO [19] and the recipe from DeXtreme [7] in the nominal source simulator, then deploy it in the target domain to record T-step trajectories $\tau = (x_0, a_0, x_1, \ldots, a_{T-1}, x_T)$. At physics interval h and time t, state $x_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{p}_t, R_t, \mathbf{v}_t, \boldsymbol{\omega}_t)$ records hand joint positions/velocities and object position, orientation, linear velocity, and angular velocity in world coordinates; R_t maps body to world, and a_t is the held joint command.

For each transition in τ, teacher-forced replay evaluates the nominal substep map $\bar{x}_{t+1} = F_h(x_t, a_t)$ from each recorded state and command. Comparing it with the recorded x_{t+1} isolates the one-step model discrepancy. Using nominal object mass m and body inertia I_B, we derive angular momentum $\mathbf{L}_{t+1} = R_{t+1} I_B R_{t+1}^\top \operatorname{Log}(R_{t+1} R_t^\top)^\vee / h$ from pose changes; $\operatorname{Log}(\cdot)^\vee$ returns a rotation vector, and $\bar{\mathbf{L}}_{t+1}$ uses the replayed $\bar{R}_{t+1}$ with the same R_t. The linear/angular momentum label in the starting object frame is

$$\mathbf{y}_t = \begin{bmatrix} \frac{m}{h} R_t^\top (\mathbf{p}_{t+1} - \bar{\mathbf{p}}_{t+1}) \\ R_t^\top (\mathbf{L}_{t+1} - \bar{\mathbf{L}}_{t+1}) \end{bmatrix}. \tag{1}$$

Contact-conditioned object residual. We represent geometry and contacts on $M = 128$ surface anchors, selected by farthest-point sampling (FPS) from area-uniform candidates. Anchor positions $\mathbf{r}_i$ and nominal contact positions $\mathbf{x}_c$, with outward unit normals $\mathbf{n}_i$ and $\boldsymbol{\nu}_c$, share the pre-step object frame centered at the object's center of mass. We splat each contact c using weights w_{ic} determined by normal alignment and tangent-plane distance ℓ_{ic}, with bandwidth σ proportional to anchor spacing:

$$\ell_{ic} = \|(I - \boldsymbol{\nu}_c \boldsymbol{\nu}_c^\top)(\mathbf{r}_i - \mathbf{x}_c)\|, \qquad w_{ic} = e^{-\ell_{ic}^2/(2\sigma^2)} [\mathbf{n}_i^\top \boldsymbol{\nu}_c]_+^2, \qquad \bar{w}_{ic} = \frac{w_{ic}}{\sum_j w_{jc} + \epsilon}. \tag{2}$$

where the kernel mass $\rho_i = \sum_c \bar{w}_{ic}$ determines activation $\chi_i = \min(\rho_i, 1)$.

The input ϕ contains eight object-frame vector channels $\mathbf{v}_{i\ell}$ ($\ell = 1, \ldots, 8$, type-1) per anchor: position, normal, mean relative contact velocity, sums of impulses and their unit directions, unit gravity, and nominal next linear/angular velocities. Four invariant scalar channels (type-0) encode χ_i, mean separation, ρ_i, and friction; goals and raw actions are excluded. The tokenizer combines the 36 upper-triangular Gram entries of $V_i^\top V_i$, where $V_i = [\mathbf{v}_{i1}, \ldots, \mathbf{v}_{i8}]$, with the scalars into a 40-dimensional token. These pairwise dot products between the vector channels remain unchanged under a common rotation [20], yielding a rotation-invariant token. After fixed z-score normalization and embedding, a PointNet++-style branch [21] adds shape context from invariant distances and dot products.

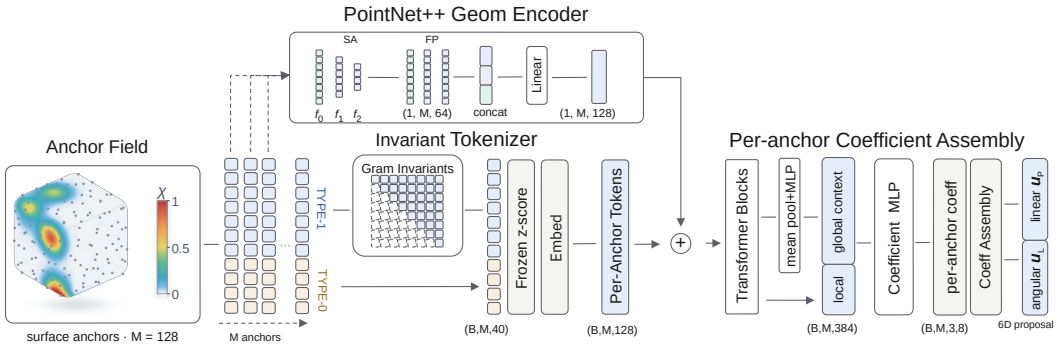


Figure 2: **Contact-Conditioned Object Residual Model.** Geometry and contact features produce a normalized proposal $\mathbf{u} = [\mathbf{u}_P; \mathbf{u}_L]$; scaling and contact bounds yield the applied correction $\hat{\mathbf{y}}$. Here B is batch size; SA/FP denote set abstraction/feature propagation, and f_0, f_1, f_2 are geometry features at three resolutions.

Two Transformer blocks [22] contextualize the anchor tokens through self-attention across anchors, producing per-anchor features $\mathbf{z}_i$. Mean pooling followed by an MLP yields a global context $\mathbf{e}$ summarizing the interaction. A coefficient MLP maps $[\mathbf{z}_i; \mathbf{e}]$ to scalar mixing weights that determine the strength and sign of each input vector's contribution to the residual. Three sets of eight coefficients per anchor form the vectors below, with normalization across anchors absorbed into $\tilde{c}_{i\ell}^{(j)}$:

$$\mathbf{b}_i^{(j)} = \sum_{\ell=1}^{8} \tilde{c}_{i\ell}^{(j)} \mathbf{v}_{i\ell}, \quad j = 1, 2, 3, \qquad \mathbf{u} = g_o(\phi) = \begin{bmatrix} \sum_i \mathbf{b}_i^{(1)} \\ \sum_i \left[\frac{\mathbf{r}_i}{r_o} \times \mathbf{b}_i^{(2)} + \mathbf{b}_i^{(3)} \right] \end{bmatrix}. \tag{3}$$

Here $r_o = \max_i \|\mathbf{r}_i\|$ scales the lever arm. The first combination forms the linear block; the other two form a lever-arm moment and a direct angular term. Invariant coefficients and vector assembly make the proposal rotation equivariant.

Residual Fitting and Policy Transfer. To obtain a momentum correction from $\mathbf{u}_t = g_o(\phi_t)$, we first apply fixed linear/angular label root-mean-square (RMS) scales from pooled training data. Nominal contacts then constrain this correction: we cap each block's norm in proportion to summed contact-impulse magnitudes to limit departures from nominal dynamics, and zero the entire correction when $\sum_i \chi_i \leq 0.5$ to avoid "ghost" corrections without effective contact. We fit this constrained output $\hat{\mathbf{y}}_t = [\Delta\mathbf{P}_t; \Delta\mathbf{L}_t]$ to the one-step labels in (1) while regularizing the raw proposals:

$$\mathcal{L} = \mathbb{E}_t \left[\gamma_t \|\hat{\mathbf{y}}_t - \mathbf{y}_t\|_{Q_t}^2 + \lambda_u \|\mathbf{u}_t\|_2^2 \right] + \lambda_s \mathbb{E}_{(t,t+1)} \left[\|\mathbf{u}_{t+1} - \mathbf{u}_t\|_2^2 \right]. \tag{4}$$

Here $\|\mathbf{d}\|_{Q_t}^2 = \mathbf{d}^\top Q_t \mathbf{d}$, γ_t is sample confidence, and diagonal Q_t uses inverse squared label RMS weights for each object/physics setting. On teacher-forced data, λ_u penalizes magnitude, while λ_s encourages smoothness between adjacent predictions. During residual training, we also vary anchor layouts to encourage learning from surface geometry rather than memorizing anchor positions.

After each nominal substep, mass and endpoint inertia convert the correction, rotated into world coordinates, into velocity increments and first-order pose updates. PPO fine-tunes the nominal task policy into π_{ft} with fixed anchor layouts and frozen g_o; only π_{ft} transfers.

3 Experiments

Setup. We evaluate MuJoCo/MJWarp policies in Genesis, fitting separate residuals for Allegro Hand and Wuji Hand. Identification uses reorientation trajectories on ten regular and ten irregular shapes, including YCB [23] and Google Scanned Objects [24] meshes (Fig. 3). We collect 100 episodes per object in each of five mass/friction settings, totaling 10,000 episodes per hand. Reorientation tests three primitives and two unseen shapes (Table 1). For cross-task evaluation, we freeze the residual trained on reorientation data and then fine-tune rotation policies on cylinder, sphere, pear, apple, lemon, and strawberry with task-specific geometries.

Table 1: **Reorientation Transfer Performance.** Reorientation transfer in Genesis, measured by goals completed per episode. Pear and ladybug (†) are unseen shapes.

	Allegro					Wuji				
Method	Cube (↑)	Sphere (↑)	Cylinder (↑)	Pear† (↑)	Ladybug† (↑)	Cube (↑)	Sphere (↑)	Cylinder (↑)	Pear† (↑)	Ladybug† (↑)
Base	12.07	23.39	7.00	1.68	5.78	10.32	24.29	7.79	3.16	5.28
NoAdapt	$12.13_{\pm 0.07}$	$22.53_{\pm 0.24}$	$7.64_{\pm 0.18}$	$2.13_{\pm 0.04}$	$6.45_{\pm 0.08}$	$10.09_{\pm 0.14}$	$19.75_{\pm 0.78}$	$7.34_{\pm 0.54}$	$3.81_{\pm 0.20}$	$5.69_{\pm 0.63}$
DeXtreme DR [7]	$7.34_{\pm 0.27}$	$15.66_{\pm 0.35}$	$4.61_{\pm 0.25}$	$3.94_{\pm 0.20}$	$3.29_{\pm 0.16}$	$5.41_{\pm 0.03}$	$10.96_{\pm 0.23}$	$4.91_{\pm 0.33}$	$3.84_{\pm 0.42}$	$5.04_{\pm 0.07}$
SysID	$13.09_{\pm 0.13}$	**$24.96_{\pm 0.13}$**	$9.20_{\pm 0.13}$	$4.78_{\pm 0.43}$	$6.68_{\pm 0.62}$	$10.06_{\pm 0.34}$	$28.04_{\pm 0.52}$	$7.82_{\pm 0.29}$	$5.98_{\pm 0.28}$	**$6.90_{\pm 0.21}$**
Random Residual	$11.00_{\pm 0.03}$	$23.10_{\pm 0.16}$	$7.49_{\pm 0.26}$	$2.39_{\pm 0.30}$	$6.27_{\pm 0.42}$	$9.54_{\pm 0.86}$	$21.89_{\pm 1.74}$	$5.84_{\pm 0.07}$	$2.20_{\pm 0.26}$	$5.49_{\pm 0.61}$
Ours	**$15.35_{\pm 0.16}$**	$24.42_{\pm 0.29}$	**$11.53_{\pm 0.40}$**	**$8.02_{\pm 0.30}$**	**$6.72_{\pm 0.42}$**	**$12.26_{\pm 0.13}$**	**$32.63_{\pm 0.06}$**	**$10.73_{\pm 0.26}$**	**$12.03_{\pm 0.40}$**	$6.38_{\pm 0.71}$

Evaluation protocol. Base is the original policy; NoAdapt controls for additional training under the original task recipe. Fine-tuning uses matched hand–object–task initializations and approximately 100M additional transitions. Baselines include task-specific SysID, fitted to target transitions, and DR using DeXtreme's reorientation [7] or HORA's rotation recipe [8]; rotation also includes HORA's deployable second-stage policy. To isolate learned correction from perturbation effects, reorientation also includes Random Residual: Gaussian object kicks with fixed means and variances from learned-kick rollouts and identical contact gating and impulse bounds.

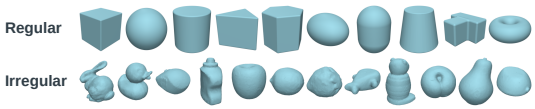


Figure 3: **Object Set.** Both hands share an object set spanning regular primitives (top) and irregular shapes (bottom) for residual identification and evaluation.

Evaluation runs at 30 Hz; reorientation counts goals over 320 control steps per episode, requiring orientation error below 0.4 rad for two consecutive steps. For rotation, we average per-object median accumulated signed world-z rotation (Rot) and mean time-to-fall (TTF) over six objects per seed. Tables 1–2 show mean $\pm$ sample standard deviation over two fine-tuning seeds (200 episodes/seed; 100 for Wuji rotation).

Reorientation. Ours achieves the highest mean in eight of ten hand–object settings and outperforms Base, NoAdapt, DeXtreme DR, and Random Residual throughout (Table 1). In the remaining two, it stays within 0.54 goals per episode of SysID. These gains extend to unseen shapes and exceed those from additional training or random perturbations, supporting the utility of learned contact corrections.

Table 2: **In-Hand Rotation Transfer Performance.** Rotation transfer in Genesis, evaluated by signed world-z rotation (Rot) and time-to-fall (TTF), averaged across six objects. Residual is trained on reorientation data and frozen for rotation fine-tuning.

	Allegro (20 s)		Wuji (60 s)	
Method	Rot (rad) (↑)	TTF (s) (↑)	Rot (rad) (↑)	TTF (s) (↑)
Base	3.60	3.28	21.05	42.20
NoAdapt	$19.13_{\pm 4.11}$	$9.60_{\pm 1.32}$	$9.33_{\pm 1.85}$	$24.62_{\pm 1.98}$
DR	$0.74_{\pm 0.11}$	$1.00_{\pm 0.12}$	$10.28_{\pm 1.99}$	$26.27_{\pm 3.68}$
HORA [8]	$7.95_{\pm 6.03}$	$6.18_{\pm 1.91}$	$23.27_{\pm 0.19}$	**$58.28_{\pm 2.14}$**
SysID	$23.48_{\pm 0.45}$	**$13.21_{\pm 0.22}$**	$10.78_{\pm 6.92}$	$29.96_{\pm 10.97}$
Ours	**$24.25_{\pm 2.71}$**	$12.28_{\pm 0.64}$	**$32.28_{\pm 1.76}$**	$57.03_{\pm 0.42}$

Cross-task Transfer. Reusing reorientation residuals yields the highest mean Rot on both hands, with TTF at 93% and 98% of the best baseline for Allegro and Wuji, respectively (Table 2). This supports cross-task reuse of the corrections for both rotational progress and object retention, without residual training on rotation data.

4 Conclusion and Future Work

We presented an object-centric contact residual for policy learning in an augmented physics simulator. Sim-to-sim experiments on Allegro and Wuji show improved transfer across object geometries, including unseen shapes, and reuse from reorientation to rotation, supporting applicability across hand embodiments and tasks. Future work will include hardware experiments, architecture ablations, and multi-step prediction tests to assess how representation and parameter count affect prediction accuracy and policy transfer.

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